

SARDAR VALLABHBHAI NATIONAL INSTITUTE OF TECHNOLOGY

Big Data analytics (AI358)

Lab Assignment - 10 (Python)

Exercise 3: Time-Series & Subplots

Goal: Practice the "Small Multiples" design dimensionality principle.

- **Dataset:** Generate a synthetic dataset or use `seaborn.load_dataset('flights')`.
- **Tasks:**
 1. Using `plt.subplots(2, 1)`, create two stacked charts.
 2. **Top Chart:** A line graph showing the trend of passengers over time.
 3. **Bottom Chart:** A bar graph showing the monthly average passengers.
 4. **Requirement:** Ensure both charts share the same X-axis (Time) and have clear annotations.

Exercise 4: Multivariate Analysis (Heatmaps)

Goal: Understand correlation matrices, similar to identifying patterns in a Modern Data Platform.

- **Tasks:**
 1. Use a dataset with at least 5 numerical variables.
 2. Calculate the correlation matrix using `df.corr()`.
 3. **Visualization:** Use `plt.imshow()` or `sns.heatmap()` to visualize the correlations.
 4. **Requirement:** Use a **diverging color map** (e.g., 'RdBu') to clearly distinguish between positive and negative correlations.